

ECON696 - Independent Study

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Motivation

As artificial intelligence tools rapidly expand across industries, concern is growing about their impact on the labor market, especially for those starting their careers. A common fear is that AI may be automating tasks traditionally performed by entry-level workers, effectively narrowing the bottom rung of the career ladder.

Entry-level positions are often the first point of access into an occupation or industry, particularly for recent graduates and those switching careers. If these positions are declining disproportionately in AI-exposed fields, it could signal growing barriers to entry for large segments of the workforce. This study investigates that possibility by measuring whether the share of entry-level job openings has declined more sharply in AI-exposed occupations than in others over the past decade.

Data

This study uses Lightcast job postings data from 2014 to 2024. Each job posting contains rich data, including:

- Occupation code
- Posting date
- Minimum required education and experience
- Skills listed in the job description

The dataset represents real-time job postings across industries in the US, capturing employer demand trends over time. It covers approximately 92% of all job postings in the US. To ensure consistency, only occupations that appeared every year from 2014 to 2024 and had a sufficient number of postings were included.

Methodology

The main idea of this independent study is to track the share of postings in each occupation that qualify as entry-level over time, and then compare the trends between occupations that are highly exposed to AI and those that are not.

The analysis is structured around four main steps:

1. Define entry-level job postings:

A job posting is considered *entry-level* if it meets the following criteria:

- Requires 1 year or less of experience OR seniority level is junior and year of experience has a null value
 - Does not require graduate education (requires at most a bachelor's degree or equivalent; specifically: "No Education Listed", "High school or GED", "Associate degree", or "Bachelor's Degree")
2. Compute entry-level share per occupation-year

For each occupation (based on 5-digit SOC 2021 codes) and each year from 2014 to 2024, the entry-level share is calculated as:

$$\text{Entry-Level Share} = \frac{\text{Entry-Level Postings}}{\text{Total Postings}}$$

3. Define AI Exposure per Occupation:

I measure occupation-level exposure to generative AI using the GPT-4 exposure score (`gpt4_beta`) provided by the authors of *Canaries in the Coal Mine* (Brynjolfsson et al., 2025). This measure builds on task-level mappings from Eloundou et al. (2023) and Handa et al. (2023), aggregating predicted GPT-4 applicability across tasks within each occupation. The exposure score is time-invariant and varies only at the occupation level. Higher values indicate greater potential exposure to GPT-4-class models.

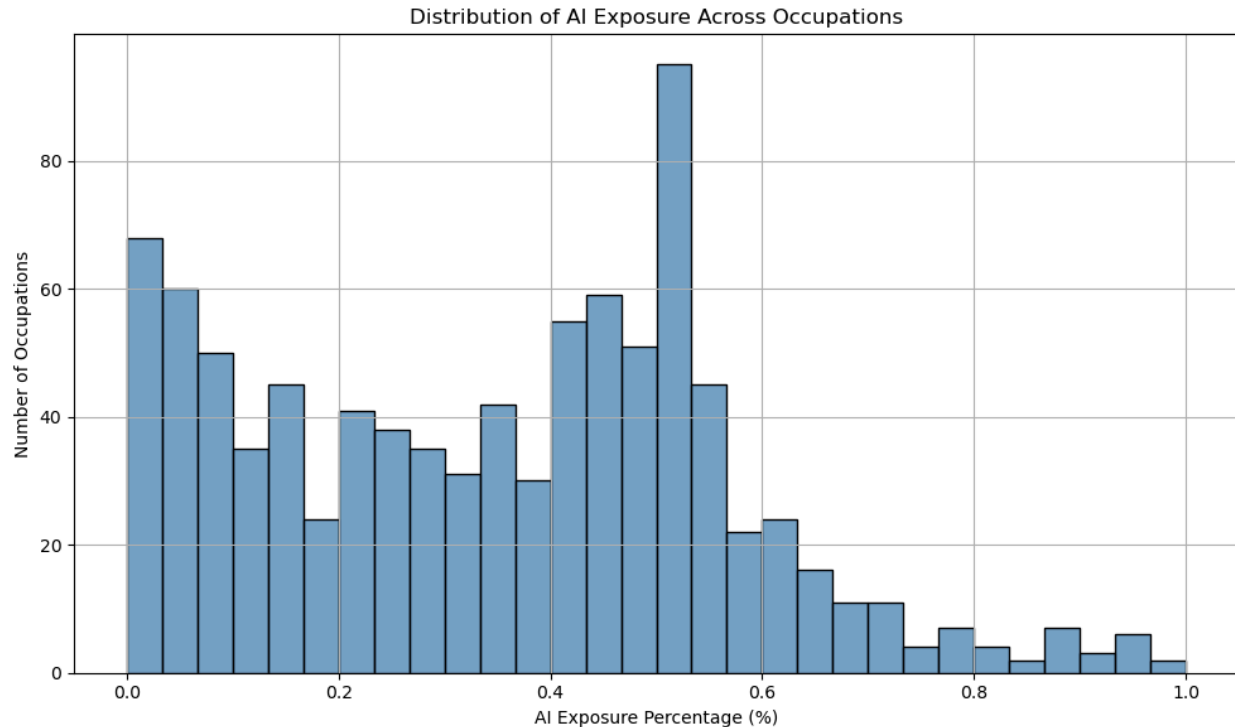
4. Seasonality adjustment, weighting, indexing, and visualization

To remove seasonal fluctuations in hiring, I residualize the entry-level share by regressing it on month-of-year fixed effects (11 month dummies) and use the residuals as a seasonally adjusted entry-level share.

When aggregating across occupations, I weight observations using fixed occupation weights based on each occupation's share of total postings in 2021, which holds the occupational composition constant over time and accounts for differences in job volume across occupations.

Occupations are then grouped into low, medium, and high GPT-4 exposure terciles based on `gpt4_beta`. For each exposure group and month, I compute a weighted average of the seasonally adjusted entry-level share.

Finally, I construct an index by normalizing each exposure group's weighted monthly series to October 2022 = 100, allowing for direct comparison of trends across exposure groups.



Findings

The figure above shows the distribution of GPT-4 exposure scores across 5-digit SOC occupations. Exposure varies substantially across occupations, with a right-skewed distribution: many occupations exhibit low to moderate exposure, while a smaller set of occupations has very high exposure to GPT-4-class models. This heterogeneity motivates grouping occupations into terciles of AI exposure (low, medium, and high) for subsequent analysis, allowing for comparisons across broad exposure categories rather than relying on a single continuous measure.

Below are top 20 occupations ranked by AI exposure rate:

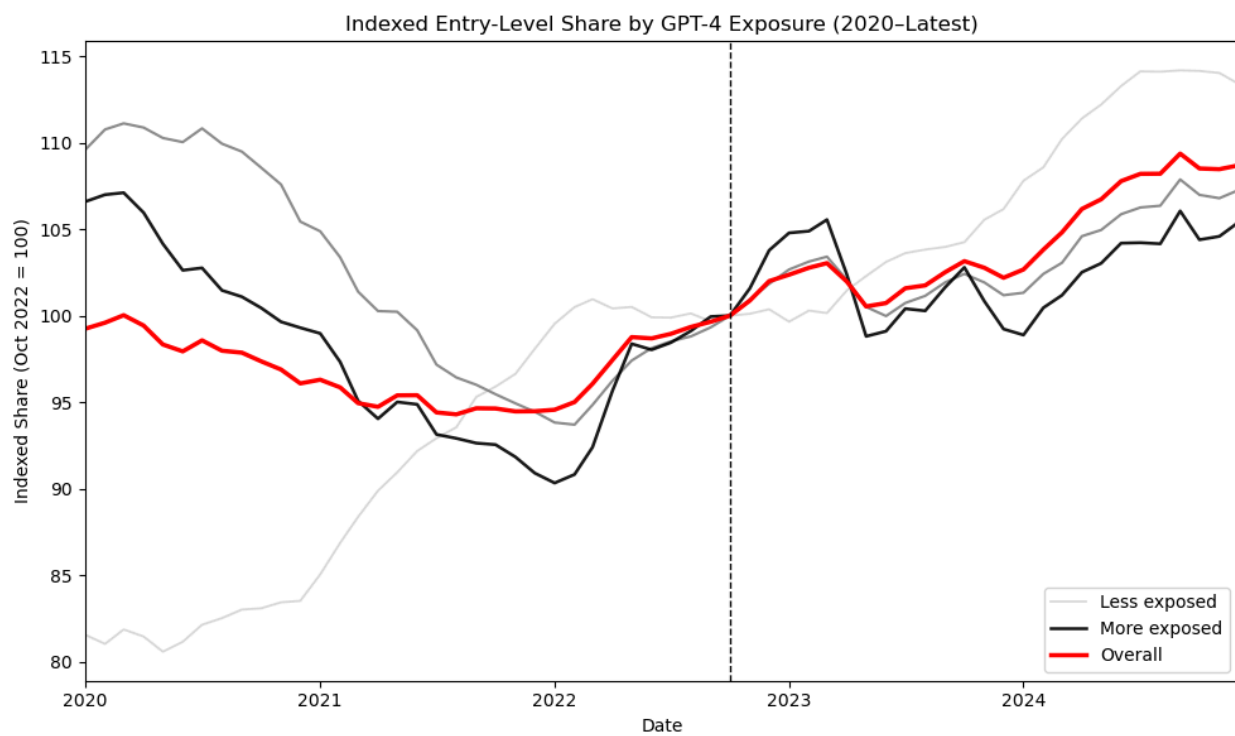
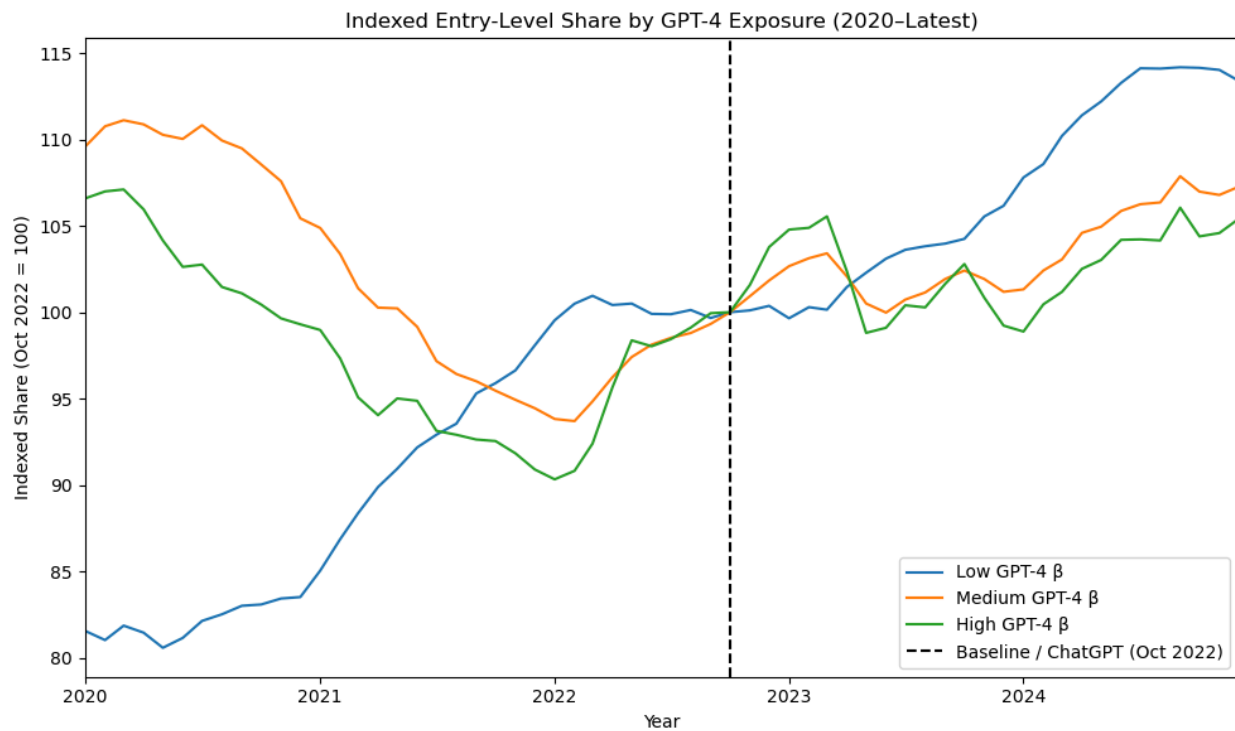
The occupations with the highest GPT-4 exposure scores include Mathematicians, Blockchain Engineers, Computer Programmers, Web Developers, Database Administrators, and Software Developers, as well as language-focused occupations such as Court Reporters, Proofreaders, Translators, and Writers. These roles tend to involve tasks, such as coding, data manipulation, mathematical reasoning, and text generation or processing, that closely align with the demonstrated capabilities of GPT-4 models. Notably, high exposure is not limited to traditional technology occupations, but also extends to clerical and communication-intensive jobs, such as Correspondence Clerks and Telephone Operators, highlighting the broad potential reach of generative AI across both technical and non-technical occupations.

A	B	C
O*NET-SOC	Title	gpt4_beta
15-2021.00	Mathematicians	1
15-1299.07	Blockchain Engineers	0.970588235
27-3092.00	Court Reporters and Simultaneous Captioners	0.964285714
43-9081.00	Proofreaders and Copy Markers	0.954545455
43-4021.00	Correspondence Clerks	0.952380952
15-1242.00	Database Administrators	0.944444444
15-1243.01	Data Warehousing Specialists	0.944444444
15-1251.00	Computer Programmers	0.941176471
15-1254.00	Web Developers	0.931034483
15-1299.01	Web Administrators	0.914285714
43-2021.00	Telephone Operators	0.90625
27-3091.00	Interpreters and Translators	0.882352941
27-3043.00	Writers and Authors	0.875
27-3043.05	Poets, Lyricists and Creative Writers	0.875
15-1252.00	Software Developers	0.868421053
15-2099.01	Bioinformatics Technicians	0.868421053
15-1253.00	Software Quality Assurance Analysts and Testers	0.866666667
31-9094.00	Medical Transcriptionists	0.866666667
15-1243.00	Database Architects	0.86
43-3051.00	Payroll and Timekeeping Clerks	0.840909091

Figure 2 plots the indexed share of entry-level job postings by GPT-4 exposure rate from 2020 onward, with October 2022 normalized to 100.

Prior to 2022, entry-level shares in low-exposure occupations rose steadily, while medium- and high-exposure occupations experienced a gradual decline, leading to convergence across exposure groups by late 2022. Following the introduction of ChatGPT, low-exposure occupations exhibit a sustained increase in the entry-level share, rising to roughly 10–15 percent above the October 2022 baseline by 2024, while medium- and high-exposure occupations show more muted growth and greater volatility.

Overall, the figure suggests that entry-level postings have expanded more rapidly in occupations with lower GPT-4 exposure relative to those with higher exposure. While entry-level opportunities in AI-exposed occupations do not collapse after 2022, their growth appears slower compared to less exposed occupations. This pattern is consistent with the possibility that generative AI disproportionately substitutes for junior-level tasks in highly exposed roles or raises skill requirements at entry, potentially limiting access to early-career positions in AI-intensive occupations.



The figure above raises an important question: if entry-level opportunities are growing more slowly in AI-exposed occupations, are employers simultaneously raising skill requirements within those occupations? To investigate this mechanism, the next stage of the project will examine changes in skill demand for entry-level postings in AI-exposed versus less exposed occupations.

Building on the framework in *Upskilling: Do Employers Demand Greater Skill When Technology Advances?* (Hershbein and Kahn, 2018), I plan to analyze whether AI exposure is associated with increases in required skills among entry-level job postings. Specifically, I will focus on three dimensions of skill demand observable in the Lightcast data:

Skill intensity

I will track changes over time in the number of distinct skills listed per entry-level job posting, comparing trends across low-, medium-, and high-GPT-4 exposure occupations. An increase in skill counts in highly exposed occupations would be consistent with employers raising hiring standards in response to AI adoption.

Skill composition

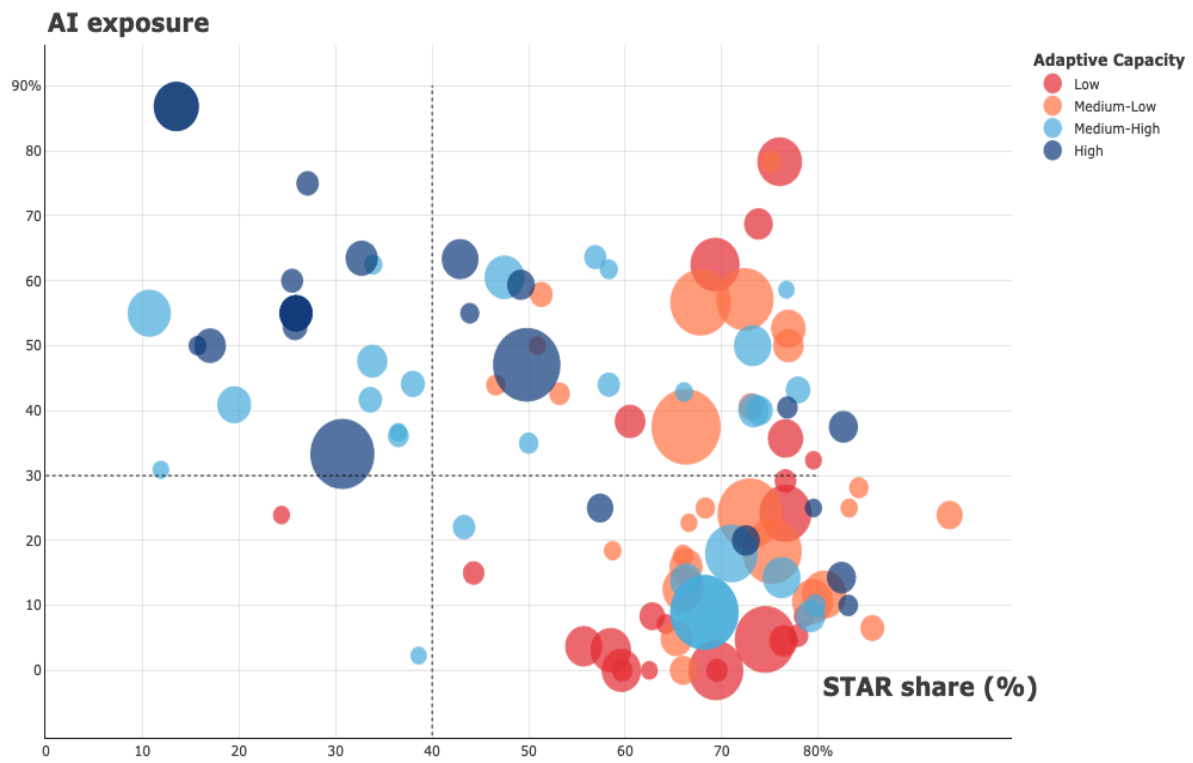
Using skill categories provided by Lightcast, I will examine whether AI-exposed occupations show disproportionate growth in:

- technical and analytical skills,
- software, data, or AI-related skills, and
- complementary “human” skills

Credential and experience proxies

I will assess whether entry-level postings in AI-exposed occupations increasingly require higher education levels, specific certifications, or named technologies.

Appendix



Source: Opp@work analysis using Brookings adaptive capacity estimates